

Transformer Image Recognition System Based on Deep Learning

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Abstract—Due to the variety of transformer models and the complexity of the image, the traditional method based on machine learning for the artificial design features cannot be large-scale and accurate identification. This paper proposes a transformer image recognition system based on deep learning, which can acquire the target results directly by "end-to-end" learning from the original acquired images. In order to realize the accurate classification of transformer images, this paper selects convolutional neural network algorithm to apply to transformer image recognition, constructing a transformer image recognition system based on convolutional neural network by designing the network structure and classifier of convolutional neural network algorithm. By constructing training set and testing set of the transformer image, the model is trained and its performances are tested. It is shown in the experimental results that compared with BP neural network model and SVM model, convolutional neural network has better effects, reaching 9.17% of the identification error, and realizing the accurate distinction of three types of transformers.

Keywords—deep learning; transformer; image recognition; convolutional neural network

I. INTRODUCTION

With the rapid development of computer vision technology, as well as the traditional substation gradually to the development of the smart substation, the unattended substation with no inspection has become a trend. High-definition video, infrared thermal image monitoring and other intelligent detection techniques are used to realize the unattended station [1-3]. Using embarked cameras and infrared thermal image cameras of unmanned inspection and inspection robot for rapid inspection of substation inspection technologies are also quickly applied [4-6]. The realization of automatic detection and recognition of critical power equipment in a large number of image data streams can improve the profit conversion rate instead of manual image processing.

Traditional power equipment recognition based on machine learning algorithm mainly uses the artificial design features as a feature of the image, and the representative ones are Scale invariant feature transform (SIFT) features[7], Histogram of Oriented Gradient (HOG)[8] coupled with efficient classifier in the electric power research of image classification. But the designing principles of these methods are often based on specific categories and specific types of equipment, designers need design characteristic according to different object special. Moreover, these methods are not

capable of improving the accuracy of image recognition with the increase of the number of samples.

In this paper, Convolution Neural Network (CNN) algorithm [9-12] based on deep learning is adopted to recognize transformer images of different models under the complex background of power station. A transformer image recognition system based on convolutional neural network is constructed by designing the network structure and classifier of convolutional neural network algorithm. Finally, this design is verified through specific experiments, which makes it practical and scientific, improving the recognition efficiency and accuracy of transformer image, and laying the foundation for subsequent fault diagnosis of power equipment.

II. INTRODUCTION TO OVERALL ALGORITHM

A. Algorithm description

The overall algorithm of this paper is shown in figure 1, which is divided into two parts. Part 1 is image preprocessing and part 2 is the design and training of CNN network. In the image preprocessing aspect, first cuts the picture to ensure the transformer as far as possible center, then unifies the picture size. Image cropping and image flipping methods are used to enhance the data through image rotation and then to add labels to the image dataset. The data set was divided into training set and test set. The training set was used to train the network model, and the test set was used to verify the effect of the network on the test set. Then the CNN network structure was constructed to extract the image features and classify the results. The structural parameters of CNN network are generated through the training set. After the model training is completed, the effect is verified on the test set.

B. CNN Network Structure

The design inspiration of Convolution Neural network is from visual recognition mechanism, through sparse connections between neurons. CNN contains convolution computation and has the depth of the structure of feedforward Neural Networks. CNN belongs to a kind of multilayer Neural Networks, which is mainly used to identify two dimensional characteristics [13-14]. Compared with the conventional Backward Propagation neural network, the convolution layer and down sampling layer are added based on it, and CNN can carry out translation invariant classification. The network structure of CNN is mainly composed of four parts, including convolutional layer (C layer), descending sampling layer (S

layer), full connection layer (F layer) and SoftMax classification layer. In the face of image translation, rotation and other deformation, CNN network has strong recognition ability. The specific structure of CNN network is shown in figure 2.

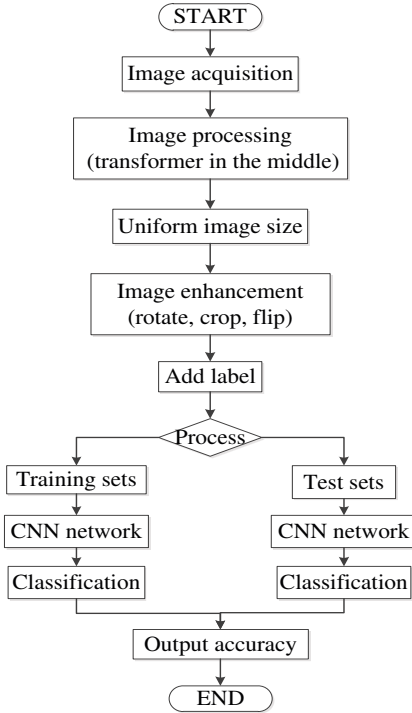


Figure1. Overall algorithm description

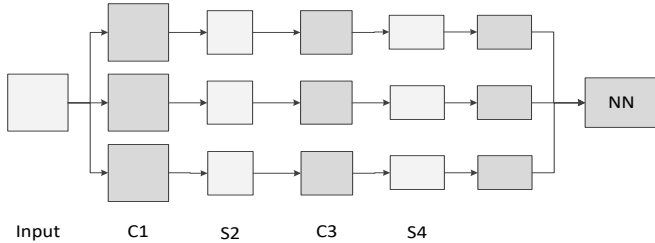


Figure2. CNN network structure

C. CNN Network Training Process

According to the essence of CNN's input and output, it can also be regarded as a nonlinear mapping. The mapping relationship between input and output can be formed by network learning mainly through the training of known models. In order to ensure the success rate of training, CNN will take the initial value operation on the network weight before the model training, so as to carry out a series of training work. The specific training process has the following two points.

1) Forward propagation stage.

In the forward propagation stage of CNN, the samples in the data set are mainly input into the network, and the output results are obtained through the combination of neuron

transformation in each layer, which is roughly the same as the forward and backward transmission mode of the traditional neural network. In the forward propagation stage, the output of the upper layer of the network is the input of the current layer. Assume the input of the current layer be $a^{[l-1]}$, and the output be $a^{[l]}$, and the relationship between these two kinds is:

$$a^{[l]} = g^{[l]}(W^{[l]}a^{[l-1]} + b^{[l]}) \quad (1)$$

In the formula: l – the number of layers; W – weight; b – offset; g – activation function

2) Back propagation stage.

When CNN completes a forward propagation, it is necessary to define an error to represent the network state after the completion of propagation. In the CNN directional propagation stage, the error is back-propagated from back to front, so that the upper neurons can update their own weights after receiving the error.

III. TRANSFORMER IDENTIFICATION CONSTRUCTION BASED ON CONVOLUTIONAL NEURAL NETWORK

A. Convolutional Neural Network Identification Construction

In the image recognition design of transformer based on convolutional neural network, the CNN network structure is mainly shown in figure 3.



Figure3. CNN network transformer image network architecture

1) Input Layer

The input layer transmits the grayscale pixel matrix of the transformer as input to the neural network. In order to ensure that the image can be transmitted into the convolutional neural network model, the size of input transformer image shall be in the standard of 32×32 .

2) Convolutional Layer

The convolution layer performs convolution operation (computation) on the input layer (for) several times by using a convolution kernel, extracts features, and combines the results into a feature matrix, and then connects to the pooling layer through a sparse connection. The size of the convolution kernel determines the local connection region. The recognition performance of CNN is directly affected by the local connection area. The convolution kernel size selected in this paper is 5×5 , and the number of convolution kernels is 6. By convolving the original 32×32 gray image, six feature maps C1 with a size of 28×28 are obtained. Similarly, the convolution layer performs a convolution operation on the 14×14 feature layer S1 and the number of convolution kernels is 16. 16 convolution operations of the feature map C2 having a size of 10×10 are obtained. The depth of the feature map is consistent with the depth of the current convolution kernel.

Actually, it is a repeated process of the linear operation of the convolution kernel and the specific formula is as follows:

$$y_{conv} = f \left(\sum_{j=0}^{j=1} \sum_{i=0}^{i=1} x_{m+i,n+j} w_{ij} + b \right) \quad (2)$$

$(0 \leq m \leq M, 0 \leq n \leq N)$

In the formula: x is the input acceptance area, which is a matrix which length is M and width is N ; w is the convolution kernel, and the length and width of the convolution kernel are j and i ; b is the bias term added to each convolution operation; y_{conv} is the output of the convolution; M, N are the length and width of the input matrix; f is the activation function of the convolution layer.

3) Pooling Layer

The down sampling layer is also called the pooling layer. Its main function is to aggregate the characteristics of different areas of an output layer, that is, to compress the results and get more important features. After the pooling layer processing, the scale of the feature graph will be reduced to a certain extent, and the number of subsequent relevant network weight parameters will be greatly reduced when the pixel points of the feature graph are reduced [15]. Maximum pooling and average pooling are the two most commonly used pooling methods. In this paper, the maximum pooling method is adopted for sampling. The sampling window of 2×2 is adopted for all samples, and the scaling factor is always set as 2, that is, the image is scaled at a step size of 2. Through the descending sampling processing of C2 in the feature layer of 10×10 , 16 feature graphs S2 with the size of 5×5 can be obtained. Similarly, the formula of the maximum pooling function is as follows:

$$f_{pool} = \text{Max}(x_{m,n}, x_{m+1,n}, x_{m,n+1}, x_{m+1,n+1}) \quad (3)$$

$(0 \leq m \leq M, 0 \leq n \leq N)$

In the formula: $x_{m,n}, x_{m+1,n}, x_{m,n+1}, x_{m+1,n+1}$ are the four values in the sampling window; M, N is the length and width of the input matrix; f_{pool} is the output of the maximum pooling. The function of the Max is to find all the maximum values in the parentheses.

4) Output Layer

After several rounds of convolutional layer and de-sampling layer processing, the input image information is abstracted into features with higher information content, which is equivalent to automatic extraction of features in the image. After feature extraction, the output layer is used for classification. When the output layer connects with the graph of the S2 layer, the connection mode is full connection. The 16 5×5 characteristic graphs of S2 layer contain a total of 400 neurons, but this study only needs to classify and identify three types of transformers, so the number of neurons in the output layer is set as 3, and there are eventually 1200 connections.

B. Classifier Design

The classifier used by SoftMax layer is SoftMax regression model. SoftMax regression, as an optimization

algorithm, turns the output of the neural network into a probability distribution, and its output tag y has k different values. The training sample is composed of m groups of input samples and output tags:

$$T = \{ (x^{(1)}, y^{(1)}), \dots, (x^{(m)}, y^{(m)}) \} \quad (4)$$

In the formula: $x^{(m)} \in \{1, 2, \dots, k\}$ is a set of input samples, $y^{(m)} \in \{1, 2, \dots, k\}$ is a classification label. After a single input sample $x^{(m)}$ passes through the classifier model, it corresponds to a recognition category j with a certain probability value, so the hypothesis function formula can be set as:

$$h_{\theta}(x^{(i)}) = \begin{bmatrix} p(y^{(i)} = 1 | x^{(i)}; \theta) \\ p(y^{(i)} = 2 | x^{(i)}; \theta) \\ \vdots \\ p(y^{(i)} = k | x^{(i)}; \theta) \end{bmatrix} = \frac{1}{\sum_{j=1}^k e^{\theta_j^T x^{(i)}}} \begin{bmatrix} e^{\theta_1^T x^{(i)}} \\ e^{\theta_2^T x^{(i)}} \\ \vdots \\ e^{\theta_k^T x^{(i)}} \end{bmatrix} \quad (5)$$

In the formula: $\theta = [\theta_1^T, \theta_2^T, \dots, \theta_k^T] \in R^{n+1}$ is usually represented as a parameter of the classifier model, where $\frac{1}{\sum_{j=1}^k e^{\theta_j^T x^{(i)}}}$ is to normalize the identified probability data to improve the accuracy of classification. Through continuous training and learning of the training set, the SoftMax model fitted the data curve through the iterative optimization method, and adjusted the reorganize value to minimize the loss function value. The loss function formula of the SoftMax classifier model is as follows:

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m \sum_{j=1}^k 1\{y^{(i)} = j\} \log \frac{e^{\theta_j^T x^{(i)}}}{\sum_{j=1}^k e^{\theta_j^T x^{(i)}}} \right] \quad (6)$$

With the iterative optimization of the model, the parameters of SoftMax classifier are clear, making it possible to identify and classify different input samples.

IV. EXPERIMENTAL DESIGN AND RESULTS ANALYSIS

A. Experimental Data Set

The experimental data set adopted in this paper comes from the network collection, as shown in figure 4. There are three types of transformers, with a total of 1200 pictures. Each picture is annotated with a border and labeled with a category label. All images are divided into training set and test set. Before training the network with data set, images are processed by data enhancement methods such as flip, crop, scale, rotation and translation to enhance the data set.



Figure4. Experimental data set

B. Experimental Results and Analysis

When extracting features, the convolution neural network is greatly affected by the number of training sets. The CNN model is trained with different numbers of training sets. The results are as shown in Table 1. Compared with the number of different training sets, it is found that the model trained in 900 training sets has a lower error value from 12.2% to 8.86% and 5.33% in the test set than the model trained in 700 and 500 training sets, which reduced by 3.34 percentage points and 6.87 percentage points. It can be seen from Table 1 that the CNN model proposed in this paper has a larger number of transformer images on the training set and the test set and the model error is smaller.

TABLE I. COMPARISON OF RECOGNITION PERFORMANCE OF DIFFERENT NUMBER OF TRAINING SETS

Number of training sets	Training Error (%)	Prediction Error Rate (%)
500	3.67%	12.2%
700	3.14%	8.86%
900	2.56%	5.33%

The training error is the effect of different models on 900 images in the training set, and the prediction error is the effect of different models on 300 images in the test set, and the results are as shown in Table 2. It can be seen from table 2 that the recognition error of the CNN model proposed in this paper on the transformer image is far lower than that of the back propagation (BP) neural network model and Support Vector Machine (SVM) model, both in the training set and the test set, with a higher image recognition rate.

TABLE II. COMPARISON OF IDENTIFICATION PERFORMANCE RESULTS

Network Model	Training Error (%)	Prediction Error Rate (%)
BP	8.89%	13.33%
SVM	12.22%	19.67%
CNN	2.56%	5.33%

The comparison of the effects of the VGG16 convolution neural network model and the CNN network model on the training set of 900 images is as shown in Table 3. The prediction error is the effect of different models for the 300 images of the test set. It can be seen from Table 3 that the VGG16 identification error is lower than the CNN network model, but the difference is not large. Since the number of VGG16 parameters is much larger than the number of CNN networks, the CNN network speed is better than the VGG16 network at the recognition speed.

TABLE III. COMPARISON OF PERFORMANCE RESULTS OF DIFFERENT MODEL RECOGNITION

Network Model	Training Error (%)	Prediction Error Rate (%)	Time (ms/one image)
CNN	2.56%	5.33%	22
VGG16	2.11	4.67%	40

V. CONCLUSION

The comparison results show that the error rate of traditional BP neural network and SVM transformer image recognition algorithm is much higher than that of CNN

transformer image recognition algorithm. The reason for this result is that the traditional transformer image recognition algorithm used too many pre-processing methods to pre-process the recognized image before feature extraction, so that the image generally has good features. Thus, the image feature extraction can only extract good features among them.

However, the acquisition results will be greatly affected by human factors, resulting in many uncertainties in the traditional extraction methods, which will affect the recognition results and lead to the phenomenon of excessive error rate. CNN differs from the traditional method in the process of transformer image recognition and it can directly input the two-dimensional image of the transformer into the network, and then identify the original image through the neural network. This process requires less pre-processing and it reduces the impact of human factors greatly and the final error rate. According to the above analysis and data analysis results, the application of CNN in transformer identification is of practical value and has more advantages than traditional identification methods.

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REFERENCES

- [1] LIAO Yunkai. Remote Video-Monitoring and Controlling System in Substation. Techniques of Automation & Applications, 2010,29(17):104-106.
- [2] LIAN Bizhan, YANG Yang. Research and design of electric power video surveillance integration platform. Power System Protection and Control, 2011,39(6):124-130.
- [3] XIAO Dong, LIN Li. Solution Scheme of Integrated Video Surveillance Platform of Power System. Automation of Electric Power Systems, 2013,37(5):74-80.
- [4] ZHOU Lihui, ZHANG Yongsheng, SUN Yong. Development and Application of Equipment Inspection Robot for Smart Substations. Automation of Electric Power Systems, 2011,35(19):85-98.
- [5] ZUO Min, ZENG Guangping, TU Xuyan. Research on Visual Navigation of Untended Substation Patrol Robot. Acta Electronica Sinica, 2011,39(10):2464-2468.
- [6] Zhang Haibo, Yuan Kui, Zhou Qingrui. Visual navigation of a mobile robot based on path recognition. Journal of Image and Graphics, 2004,9(7): 853-857.
- [7] LI Jian, WU Jianhua, ZHANG Yuliang. Research on Suspension Recognition of Power Equipment Based on SIFT Matching. Electrical Engineering and Automation, 2011,30(4):7-9.
- [8] YANG Yonghui, LIU Changping, HUANG Lei. Application of image and video analysis in monitor system of power equipment. Journal of Computer Applications, 2010,30(1):281-284.
- [9] Ciresan, Dan; Meier, Ueli; Schmidhuber, Jürgen. Multi-column deep neural networks for image classification, 2012 IEEE Conference on Computer Vision and Pattern Recognition, pp. 3642-3649.
- [10] Justin Ker, Yeqi Bai, Hwei Yee Lee, Jai Rao, and L. P. Wang. Automated brain histology classification using machine learning, "Journal of Clinical Neuroscience, vol.66, pp. 239-245, 2019.
- [11] Justin Ker, Satya P. Singh, Yeqi Bai, Jai Rao, Tchoyoson Lim and L. P. Wang. Image Thresholding Improves 3-Dimensional Convolutional Neural Network Diagnosis of Different Acute Brain Hemorrhages on

Computed Tomography Scans, Sensors, 2019, 19, 2167; doi:10.3390/s19092167.

[12] Justin Ker, L. P. Wang, Jai Rao, and Tchoyoson Lim. Deep Learning Applications in Medical Image Analysis, IEEE Access, vol.6, pp. 9375-9389,2018.

[13] LI Junfeng, HE Shuangbo, FENG Weixia. Improved CNN based transformer image recognition technology in augmented reality environment. Modern Electronics Technique, 2018,41(7):29-32.

[14] LIU Wanjun, LIANG Xuejian, QU Chenghai. Adaptive Enhanced Convolutional Neural Network Image Recognition. Journal of Image and Graphics, 2017,22(12):1723-1736.

[15] ZHOU Junyu, ZHAO Yanming. A Survey of Convolutional Neural Networks in Image Classification and Target Detection. Computer Engineering and Applications, 2017,53(13):34-41.

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